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PRACTICE EXAM **Answer: (with solutions)**
ECE 657A - Data and Knowledge Modelling and Analysis
University of Waterloo
Term: Winter Year: 2023

Actual Exam is at... Date: Friday, April ?, 2022

Time: ?::?pm – ?::? pm (150 minutes)

Location: DC ?::?

Instructors: Prof. Mark Crowley

Lecture Section: 001

Exam Type: In person written test

Number of exam pages (includes cover page): ??
(last year as 16)

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Section _____

Instructions

1. Answer the questions in the spaces provided. If you require additional space to answer a question, the last two pages and refer to those pages in your solutions. Do not tear off the pages in the booklet.
2. Use a dark pencil or pen for your work. Additional Materials: Notes (6 pages, handwritten, double-sided) and calculator.
3. For multiple choice and true-false questions fill in the *single best option* as your answer on the bubble sheet on the final page. Corrections to the bubble sheet can use an eraser or cross out an option and add a description of the error in writing in the margins.
4. Exam invigilators will only answer questions for clarification, or if you think there is a mistake in the exam. If something is still unclear, then state your assumptions and understanding of the question on the page and answer as best you can.
5. Washroom breaks: if you need to go to the washroom, raise your hand and an invigilator will escort you to the outside of the washroom, leave your phone in your bag. Only one student can go at a time.

True or False

Answer the following questions with either True or False. Enter your answer on the **bubble sheet** at the back of your exam, use “A” for True and use “B” for False. Note: you *should* circle the correct answer on the question sheet as well for reference, this is also helpful when grading, but be sure to fill in the correct option on the bubble sheet.

1. **(2 points) (False):** Backpropagation is no longer widely used for training deep neural networks.
2. **(2 points) (True):** Backpropagation does not provide a guarantee of reaching a globally optimal solution.
3. **(2 points) (False):** Backpropagation is very difficult to parallelize to gain computational efficiency, which is why CNNs were such an important advance.
4. **(2 points) (True):** Backpropagation merely minimizes errors relative to training data, it does not do anything to help with generalization.

Short Answer...What If?

Answer the following questions with Yes or No by circling here and entering “A” for Yes or “B” for No on the bubble sheet. Then write on the exam paper a short, one or two sentences, explanation for your answer.

5. **(2 points)** For anomaly detection, if we changed the Isolation Forest algorithm to use only one tree rather than many, would it still perform well? **(No)**
6. **(3 points)** Briefly explain why: **Answer:** No, using many trees is essential for iForest since the trees themselves are grown very randomly and may not represent a good breakdown of the datapoints, so the depth property would not be meaningful to determine whether a point was an anomaly or not.
7. **(2 points)** For regression, if we changed the non-linear activation functions of a neural network to linear functions, would it still perform well? **(No)**
8. **(3 points)** Briefly explain why: **Answer:** No, nonlinear activation is an essential part of what makes NNs powerful. They need to decide between which features to use and which to discard, so these thresholds are needed. Otherwise, everything would only be complex linear regression.
9. **(2 points)** For autoencoders, if we used multiple narrow “bottleneck” layers in a row, would it still be possible to learn a good representation of the data? **(Yes)**

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10. **(3 points)** Briefly explain why: **Answer:** Yes, it would be possible, but it might be very hard to train since there is such a tight set of constraints. Then again, the encoded layers could just copy each other to keep it simple.
 11. **(2 points)** For clustering, if we change the DBScan algorithm to treat border points the same as noise points, would the algorithm still perform similarly well? **(Yes)**
 12. **(3 points)** Briefly explain why: **Answer:** Yes, probably. The border points themselves are just points that are close enough to be in the core cluster, but not surrounded by enough other points which are. They satisfy the ϵ distance criteria, but not the m neighbours within that distance criteria. There would be more noise points, but the core points would be the same.

Multiple Choice

For each of the following questions choose a single choice which best answers the question. Enter your answer on the **bubble sheet** at the back of your exam for the corresponding question number and letter. Note: you can circle the correct answer on the question sheet as well for reference, but be sure to fill in the correct option on the bubble sheet.

13. no examples, but there will be MC questions

Deep Neural Network Mad Libs

***Note:** This question was a new idea I had last year, it was a bit confusing for people and some adjustments had to be made. So this exact type of question will not be on the exam, but it has useful concepts to think about.*

Consider two Deep Neural Network components A and B which be anything including activation functions, fully connected or convolutional layers, the softmax function, a pooling or dropout calculation, etc. All we know is that A takes inputs first (it could be the original data or the output of a previous DNN component) and then B is applied to A in some way.

The following “Mad Lib” style template defines a sentence describing the relationship between A and B which is quite universal for DNN architectures.

B is a $\left(\left(\begin{array}{c} \text{fixed} \\ \text{parameterized, learnable} \end{array}\right)\right), \left(\left(\begin{array}{c} \text{linear} \\ \text{non-linear} \\ \text{randomized} \end{array}\right)\right)$ function operating on

$\left(\left(\begin{array}{c} \text{each individual} \\ \text{multiple} \\ \text{all} \end{array}\right)\right) \left(\left(\begin{array}{c} \text{internal weight(s)} \\ \text{output(s)} \end{array}\right)\right)$ of A .

Clarification Note: *that these terms relate to the types of calculations that B performs and the scope of information from A that B has access to while doing those calculations. So while a convolutional layer operates on all the outputs of the previous component at some point, each convolutional filter operator on just a subset of the outputs at a time.*

Two Examples:

A: Fully Connected Layer $\rightarrow$ B: Dropout

- | | |
|-----------|------------------------------|
| (A) fixed | (B) parameterized, learnable |
|-----------|------------------------------|

ANSWER: A
- | | | |
|------------|----------------|----------------|
| (A) linear | (B) non-linear | (C) randomized |
|------------|----------------|----------------|

ANSWER: C
- | | | |
|---------------------|--------------|---------|
| (A) each individual | (B) multiple | (C) all |
|---------------------|--------------|---------|

ANSWER: A

-
- (A) internal weight(s) (B) output(s) **ANSWER: C**

A: Max Pooling $\rightarrow$ B: Convolutional Filter

- (A) fixed (B) parameterized, learnable **ANSWER: B**
- (A) linear (B) non-linear (C) randomized **ANSWER: A**
- (A) each individual (B) multiple (C) all **ANSWER: B**
- (A) internal weight(s) (B) output(s) **ANSWER: B**

Now complete fill in the blanks for this sentence given the each of the combinations below for *A* and *B*. You can circle the answers here, but also fill in the bubble sheet at the back with the appropriate response. Each term is worth **(1 points)**.

A: Convolutional Layer $\rightarrow$ B: Average Pooling

14. (A) fixed (B) parameterized, learnable **Answer: B**
15. (A) linear (B) non-linear (C) randomized **Answer: A**
16. (A) each individual (B) multiple (C) all **Answer: B**
17. (A) internal weight(s) (B) output(s) **Answer: B**

A: Fully Connected Layer $\rightarrow$ B: Softmax

Answer: Softmax is a fixed, nonlinear function over all the outputs of the previous layer to rescale them to sum to 1.

18. (A) fixed (B) parameterized, learnable **Answer: A**
19. (A) linear (B) non-linear (C) randomized **Answer: B**
20. (A) each individual (B) multiple (C) all **Answer: C**
21. (A) internal weight(s) (B) output(s) **Answer: B**

A: Fully Connected Layer $\rightarrow$ B: ReLU

22.

(A) fixed	(B) parameterized, learnable
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Answer: A
23.

(A) linear	(B) non-linear	(C) randomized
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Answer: B
24.

(A) each individual	(B) multiple	(C) all
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Answer: A
25.

(A) internal weight(s)	(B) output(s)
------------------------	---------------

Answer: B

A: Fully Connected Layer $\rightarrow$ B: ELU

26.

(A) fixed	(B) parameterized, learnable
-----------	------------------------------

Answer: A
27.

(A) linear	(B) non-linear	(C) randomized
------------	----------------	----------------

Answer: B
28.

(A) each individual	(B) multiple	(C) all
---------------------	--------------	---------

Answer: A
29.

(A) internal weight(s)	(B) output(s)
------------------------	---------------

Answer: B

A: Fully Connected Layer $\rightarrow$ B: Sigmoid

30.

(A) fixed	(B) parameterized, learnable
-----------	------------------------------

Answer: A
31.

(A) linear	(B) non-linear	(C) randomized
------------	----------------	----------------

Answer: B
32.

(A) each individual	(B) multiple	(C) all
---------------------	--------------	---------

Answer: A
33.

(A) internal weight(s)	(B) output(s)
------------------------	---------------

Answer: B

CNN Size Calculations

Consider CNN D2 which takes input images of size 32×32 with 1 colour channels. The first layer of D2 uses 4 filters of size 5×5 , a stride of 2, and zero-padding of width 1.

34. **(3 points)** What will be the dimensions of the resulting activation map for each filter in this first layer?
- (A) 17×17
 - (B) 11×11
 - (C) 15×15 **Answer: correct**
 - (D) 19×19
35. **(3 points)** What would be the total size of the flattened output vector from each filter?
- (A) 23
 - (B) 25 **Answer: correct**
 - (C) 26
 - (D) 27
36. **(3 points)** What is the total number of weights defined for the entire activation output of this first layer? (ie. If you flattened all filters and channels into a single vector)
- (A) 289
 - (B) 900 **Answer: correct**
 - (C) 1346
 - (D) 1800

CNN Pooling Calculations

Assuming the 4 x 4 matrix below is received as input to a pooling layer somewhere in a CNN, answer the following questions:

Note: Choices are presented as 2 x 2 matrices of numbers to choose from.

10	4	2	4
12	3	1	8
6	10	7	7
13	1	5	12

37. (3 points) Which of the following outputs would be correct if we were using max pooling, with a 2 x 2 pooling filter and a stride of 2 ?

(A) **Answer: close**

12	8
14	13

(B) **Answer: distract**

14	10
15	12

(C) **Answer: correct**

12	8
13	12

(D) **Answer: distract**

12	12
20	14

38. (3 points) Which of the following outputs would be correct if we were using mean pooling, with a 2 x 2 pooling filter and a stride of 1 ?

(A) **Answer: distract**

9.25	3.5	4.75
8.75	7.25	6.75
8.5	6.75	9.75

(B) **Answer: correct**

7.25	2.5	3.75
7.75	5.25	5.75
7.5	5.75	7.75

(C) **Answer: close**

7	3.5	3.75
6.75	5.75	6.75
8.5	5.75	7.75

(D) **Answer: flipped**

7.25	7.75	7.5
2.5	5.25	5.75
3.75	5.75	7.75

DNN Network Activation Output

The following questions describe an architecture for a Deep Neural Network and ask you to choose the corresponding formulation of the total network activation using the multiple choice bubble sheet for the appropriate numbered question. The following notation will be used:

Notation:

- x_i : input data values
- d : the dimensionality of the input data x
- $n^{(c)}$: number of hidden nodes in layer (c)
- $f_A(x)$: activation function for type A where:
 $A \in \{f_R = \text{ReLU}, f_E = \text{ELU}, f_{LR} = \text{Leaky ReLU}, f_\sigma = \text{Sigmoid}, f_{\text{softmax}} = \text{Softmax}, \}$.
- $w_{ab}^{(c)}$: hidden weights for layer (c) , connecting incoming feature b from the previous layer to the hidden unit a . The bias weight for that layer, is denoted $w_0^{(c)}$.
- $a * b$: normal, pairwise multiplication of numbers a and b

Network 1

This network takes a flat 64 dimensional vector of numeric input x , with the following architecture:

- the network contains 2 fully connected, hidden layers.
- the first hidden layer, indexed by j , has 16 nodes and uses a ELU activation function.
- the second hidden layer, indexed by k , has 8 nodes and uses a ReLU activation function.
- the output layer, indexed by m , feeds the output to a Softmax activation function to produce a probability distribution over 8 possible values.

39. (3 points) Which of the following formulations is the correct one for the m -length vector output of the above Neural Network architecture? **Answer: C, some picked A, why?**

$$\text{(A):net} = f_{\text{softmax}} \left(f_R \left(\sum_{k=1}^8 w_{mk}^{(3)} * f_E \left(\sum_{j=1}^{16} w_{kj}^{(2)} * x_i + w_{j0}^{(1)} \right) + w_{k0}^{(2)} \right) \right)$$

$$\text{(B):net} = f_{\text{softmax}} \left(f_R \left(\sum_{k=1}^8 w_{mk}^{(3)} * f_R \left(\sum_{j=1}^{16} w_{kj}^{(2)} * f_E \left(\sum_{i=1}^{64} w_{ji}^{(1)} * x_i + w_{j0}^{(1)} \right) + w_{k0}^{(2)} \right) + w_{m0}^{(3)} \right) \right)$$

$$\text{(C):net} = f_{\text{softmax}} \left(\sum_{k=1}^8 w_{mk}^{(3)} * f_R \left(\sum_{j=1}^{16} w_{kj}^{(2)} * f_E \left(\sum_{i=1}^{64} w_{ji}^{(1)} * x_i + w_{j0}^{(1)} \right) + w_{k0}^{(2)} \right) + w_{m0}^{(3)} \right)$$

$$\text{(D):net} = f_{\text{softmax}} \left(\sum_{m=1}^{64} w_{mk}^{(3)} * f_R \left(\sum_{j=1}^{16} w_{kj}^{(2)} * f_E \left(\sum_{k=1}^8 w_{ji}^{(1)} * x_i + w_{j0}^{(1)} \right) + w_{k0}^{(2)} \right) + w_{m0}^{(3)} \right)$$

Network 2

This network takes a flat 100 dimensional vector of numeric input x , with the following architecture:

- the network contains 4 fully connected, hidden layers
- the first hidden layer, indexed by j , has 80 nodes and uses a Leaky ReLU activation function
- the second hidden layer, indexed by k , has 10 nodes and uses a Leaky ReLU activation function
- the third hidden layer, indexed by p , has 80 nodes and uses a Leaky ReLU activation function
- the output layer, indexed by q , feeds the output Sigmoid activation function to produce a probability distribution over 100 possible values.

40. (3 points) Which of the following formulations is the correct one for the q -length vector output of the above Neural Network architecture? **Answer: A, many people picked C or D, why?**

(A): $net_q = f_{\sigma}^{(4)} \left(\sum_{p=1}^{80} w_{qp}^{(4)} * f_{LR}^{(3)} \left(\sum_{k=1}^{10} w_{pk}^{(3)} * f_{LR}^{(2)} \left(\sum_{j=1}^{80} w_{kj}^{(2)} * f_{LR}^{(1)} \left(\sum_{i=1}^{100} w_{ji}^{(1)} * x_i + w_{j0}^{(1)} \right) + w_{k0}^{(2)} \right) + w_{p0}^{(3)} \right) \right) \right)$

(B): $net_q = f_{\text{softmax}} \left(f_{\sigma}^{(4)} \left(\sum_{p=1}^{80} w_{qp}^{(4)} * f_{LR}^{(3)} \left(\sum_{k=1}^{10} w_{pk}^{(3)} * f_{LR}^{(2)} \left(\sum_{j=1}^{80} w_{kj}^{(2)} * f_{LR}^{(1)} \left(\sum_{i=1}^{100} w_{ji}^{(1)} * x_i + w_{j0}^{(1)} \right) + w_{k0}^{(2)} \right) \right) \right) \right) \right)$

(C): $net_q = f_{\sigma}^{(4)} \left(\sum_{q=1}^{100} \left(\sum_{p=1}^{80} w_{qp}^{(4)} * f_{LR}^{(3)} \left(\sum_{k=1}^{10} w_{pk}^{(3)} * f_{LR}^{(2)} \left(\sum_{j=1}^{80} w_{kj}^{(2)} * f_{LR}^{(1)} \left(\sum_{i=1}^{100} w_{ji}^{(1)} * x_i + w_{j0}^{(1)} \right) + w_{k0}^{(2)} \right) + w_{p0}^{(3)} \right) \right) \right) \right)$

(D): $net_q = f_{\sigma}^{(4)} \left(\sum_{p=1}^{100} w_{qp}^{(4)} * f_{LR}^{(3)} \left(\sum_{k=1}^{80} w_{pk}^{(3)} * f_{LR}^{(2)} \left(\sum_{j=1}^{10} w_{kj}^{(2)} * f_{LR}^{(1)} \left(\sum_{i=1}^{80} w_{ji}^{(1)} * x_i + w_{j0}^{(1)} \right) + w_{k0}^{(2)} \right) + w_{p0}^{(3)} \right) \right) \right)$

Followup Questions

41. (3 points) If we wanted to extract the smallest useful encoding from this network and use it for another purpose as a compact representations. Which component would provide that output? **Circle the correct component here on the exam page.** (don't fill in the bubble sheet, there's too many options for this one)

net_q	$f_{\sigma}^{(4)}$	$w_{qp}^{(4)}$	$f_{LR}^{(3)}$ Answer: correct
$w_{pk}^{(3)}$	$f_{LR}^{(2)}$ Answer: ok	$w_{kj}^{(2)}$ Answer: ok	$f_{LR}^{(1)}$
$w_{j0}^{(1)}$	$w_{k0}^{(2)}$	$w_{p0}^{(3)}$	

Extra pages for answers or rough work. Please specify the question number here and the use of this page on the question page.

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